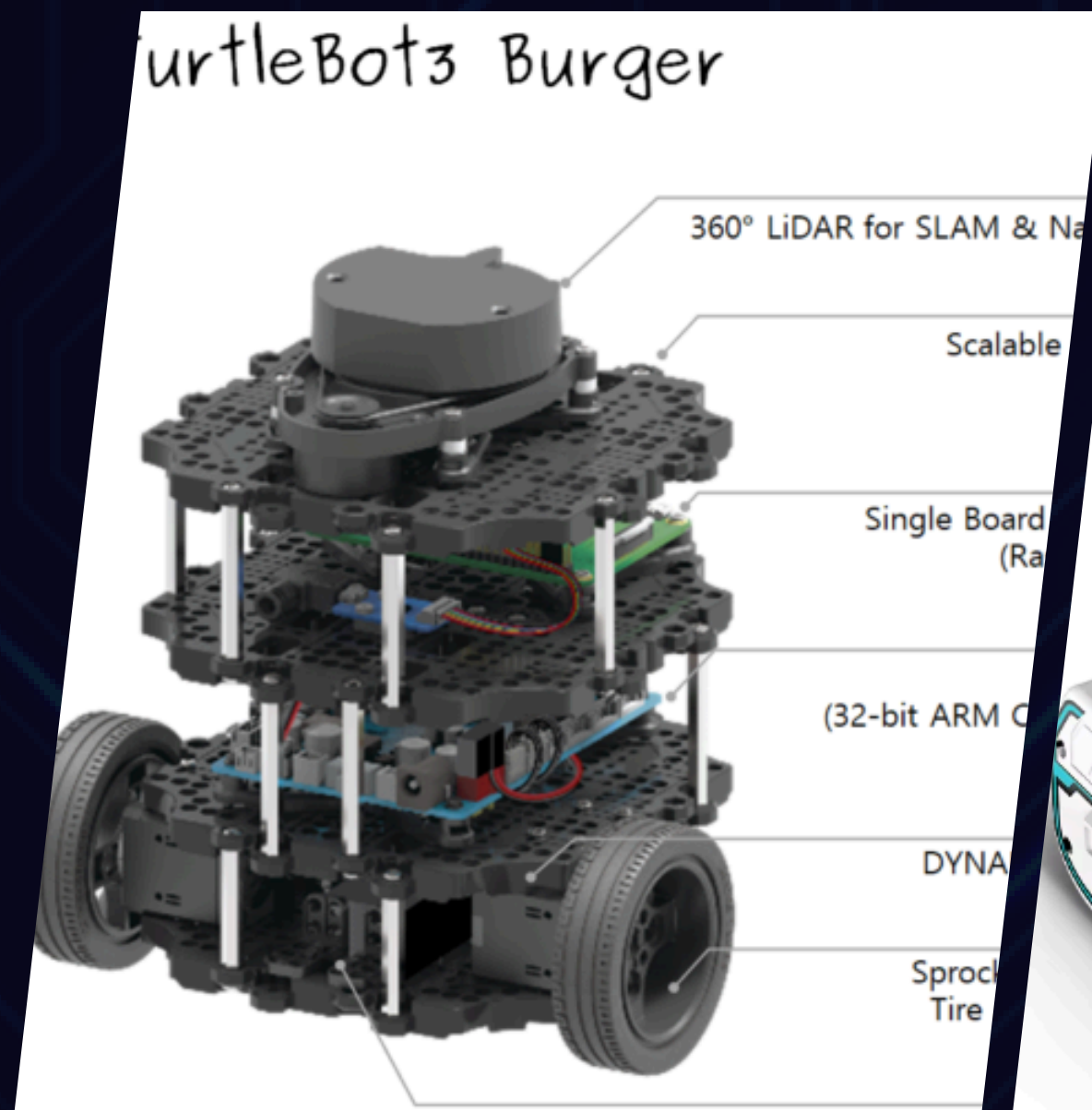


THOTH

D E S I G N P R O P O S A L



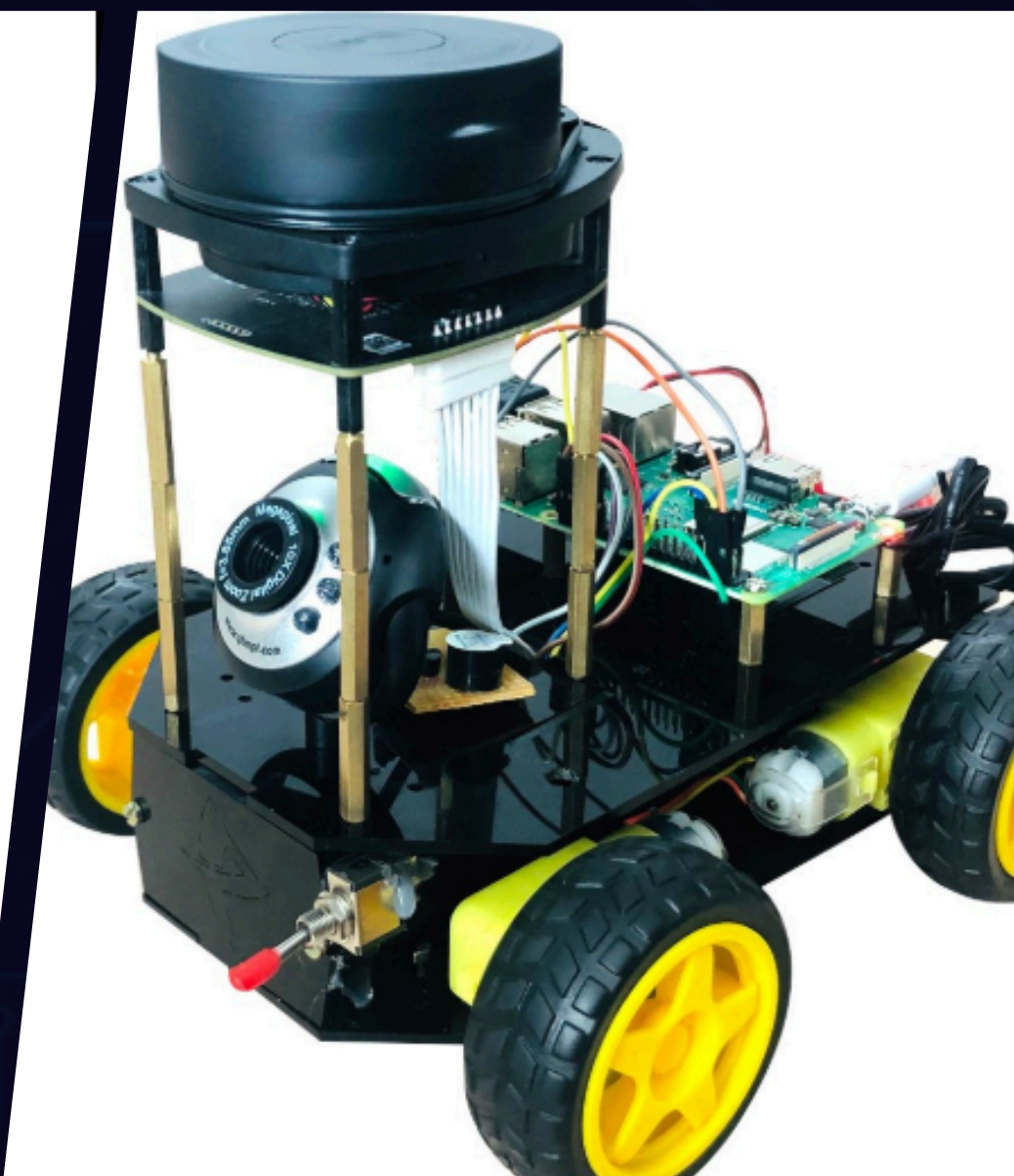
OPTIONS



TURTLEBOT BASED DESIGN



TOY BASED DESIGN



DIY NAVIGATION DESIGN

TURTLE BOT

TurtleBot is a family of open-source, low-cost mobile robots designed for research, education, and development in robotics.

Designed to:

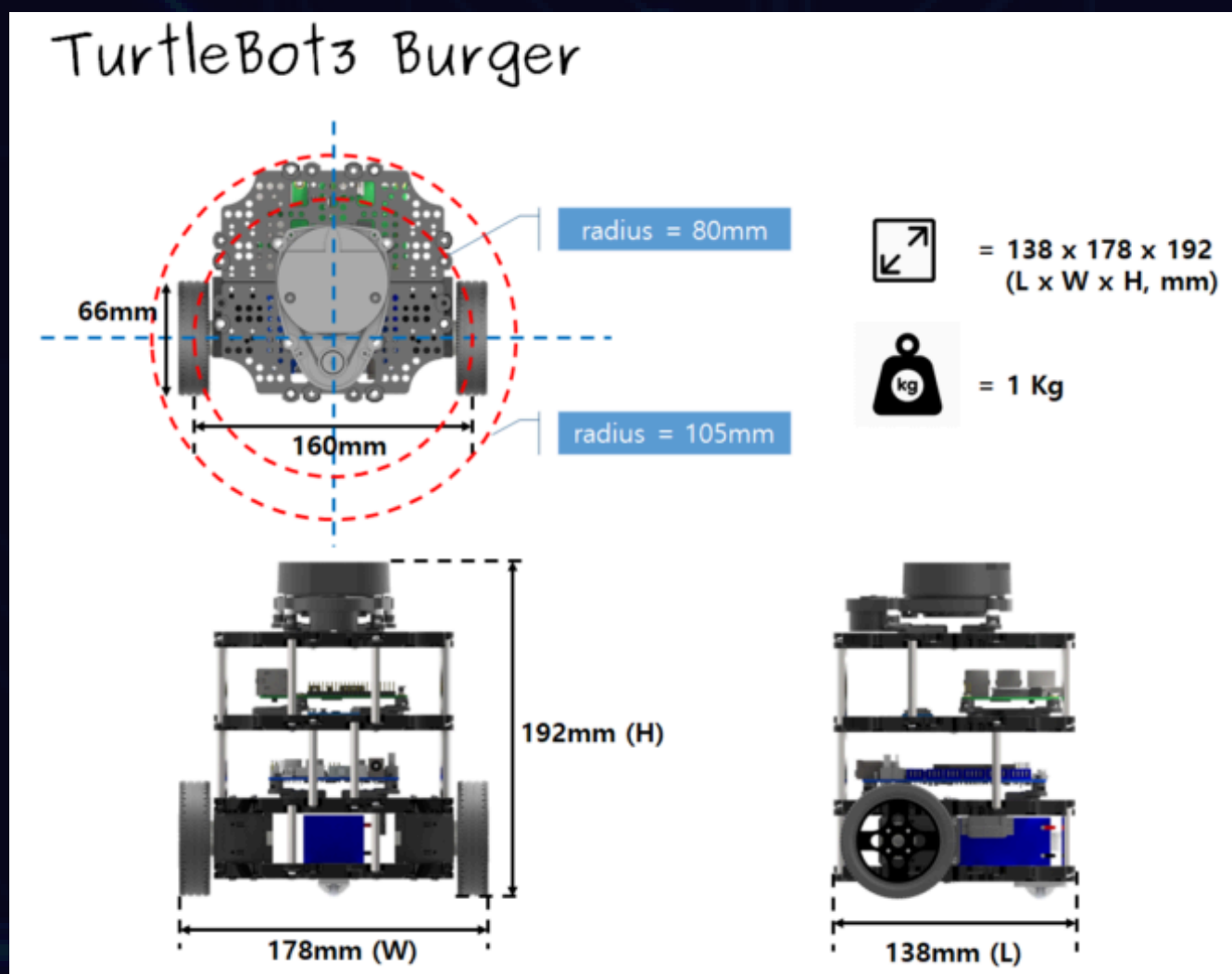
Provide a ready-to-use robotic platform so developers can focus on software, autonomy, and AI instead of building hardware from scratch.

Advantages

- ROS2 Native Support Fully compatible with modern robotics frameworks
- Reliable Navigation Stack Works well with SLAM and Nav2
- Modular Design Easy to add camera, screen, sensors
- Strong Community Support Extensive documentation and tutorials

Disadvantages

- Higher Cost Expensive compared to DIY or toy solutions
- Limited Humanoid Appearance Requires additional design (screen/head)
- Hardware Constraints Limited payload (especially Burger model)
- Less Flexibility in Low-Level Control Predefined hardware structure



**TurtleBot 3 (ALONE) costs \$783
=40,000 EGP**

TOY BASED DESIGN

A toy-based robot design involves using a commercially available robot (humanoid or wheeled) as a base platform, then modifying and extending it to support autonomous behavior and intelligent interaction.

System Concept

- Robot (Execution Unit)Handles movement (wheels, gestures, basic actions)
- Executes predefined commands
- External Computer (Laptop)Acts as the main “brain”
- Runs:
 - Decision-making logic
 - Computer vision
 - AI/LLM interaction
- Communication InterfaceSends commands such as:
 - Move forward
 - Turn
 - Perform gesture



Advantages

- Low CostSignificantly cheaper than research platforms
- Humanoid DesignNaturally more engaging for human interaction
- Reduced Mechanical ComplexityNo need for chassis design or assembly
- Focus on AI & InteractionMore time for vision, LLM, autonomy

Disadvantages

- Limited Control AccessMay require reverse engineering
- Restricted Movement CapabilitiesPredefined motions, less precision
- Lack of SensorsNo built-in LiDAR or accurate odometry
- Unreliable for NavigationNot designed for autonomous robotics

To be useful we should drop navigation and make it a simple line follower with a fixed path or else we will face problems with integrating lidar with such system !

DIY DESIGN

The DIY approach involves designing and building a custom robot platform from scratch, inspired by established research platforms such as TurtleBot.



Advantages

- Fully customizable design
- Flexible system architecture
- Scalable and extendable
- Cost can be optimized
- Strong learning opportunity
- Built on ROS 2

Disadvantages

- Longer development time
- Requires system integration effort
- Broader project scope
- Step-by-step implementation needed
- Initial setup phase required

DIY DESIGN

Components and costs:

- RPI 5 4GB → 8000 EGP
- A1 RPLIDAR 360° Laser Range Scanner → 7,500.00 EGP
- ESP32 → 300EGP
- RPI screen → 3500EGP
- FRAME+MOTORS+DRIVERS+WHEELS → 10000EGP
- USB CAMS → 1200EGP
- MISCELLANEOUS → 10000
- TOTAL :40,500egp



FEATURES

Autonomous Navigation System

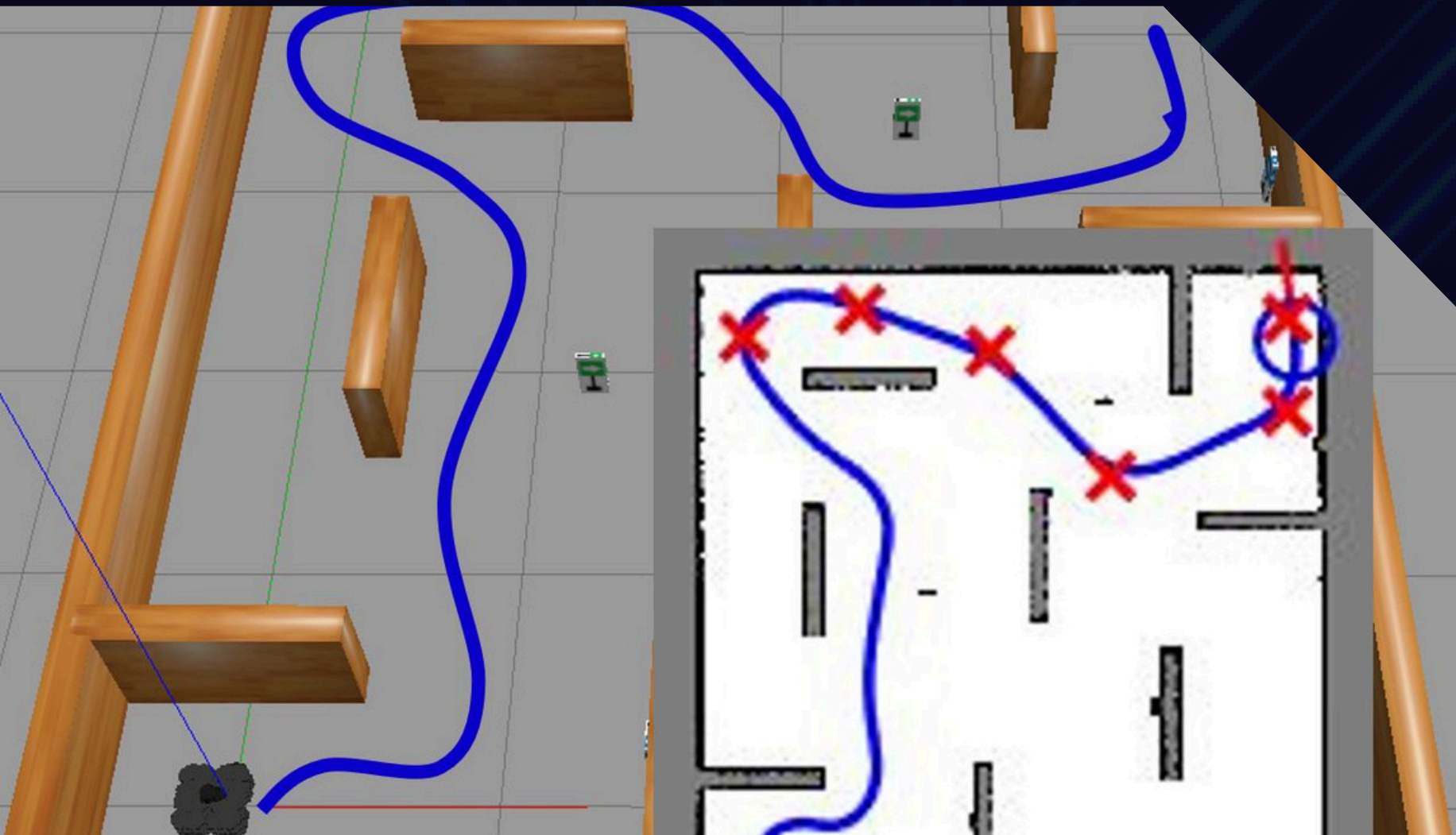
Self-guided movement between exhibits using mapped routes

Dynamic Obstacle Avoidance

Real-time detection and avoidance of people and objects

Multi-Mode Navigation

Supports guided tours, follow-me mode, and direct exhibit routing



FEATURES

Natural Language Interaction

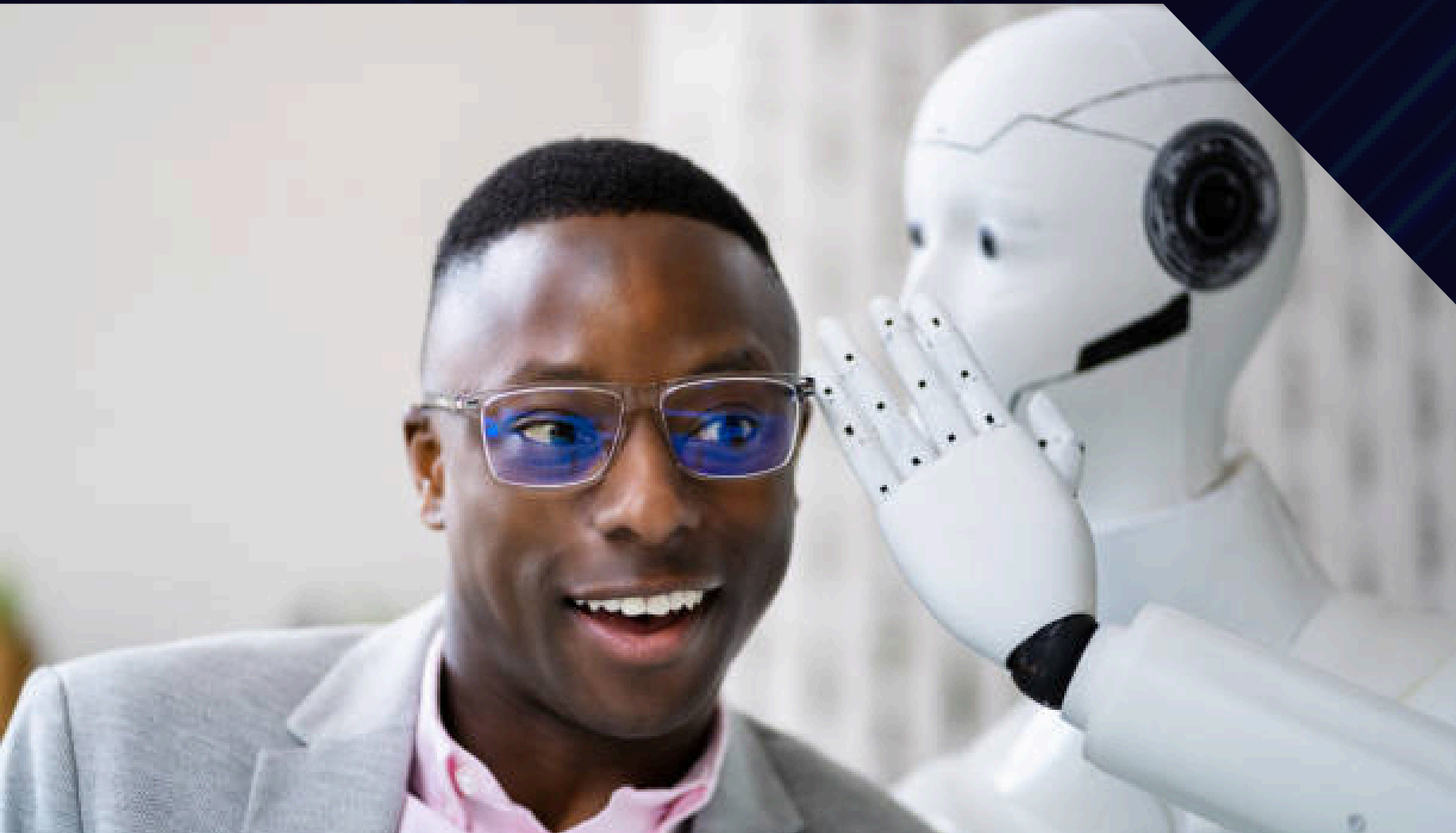
Free-form conversational Q&A using AI

Multilingual Communication

Supports multiple languages with seamless switching

Context-Aware Conversation

Remembers session context and adapts responses



FEATURES

Interactive Display System

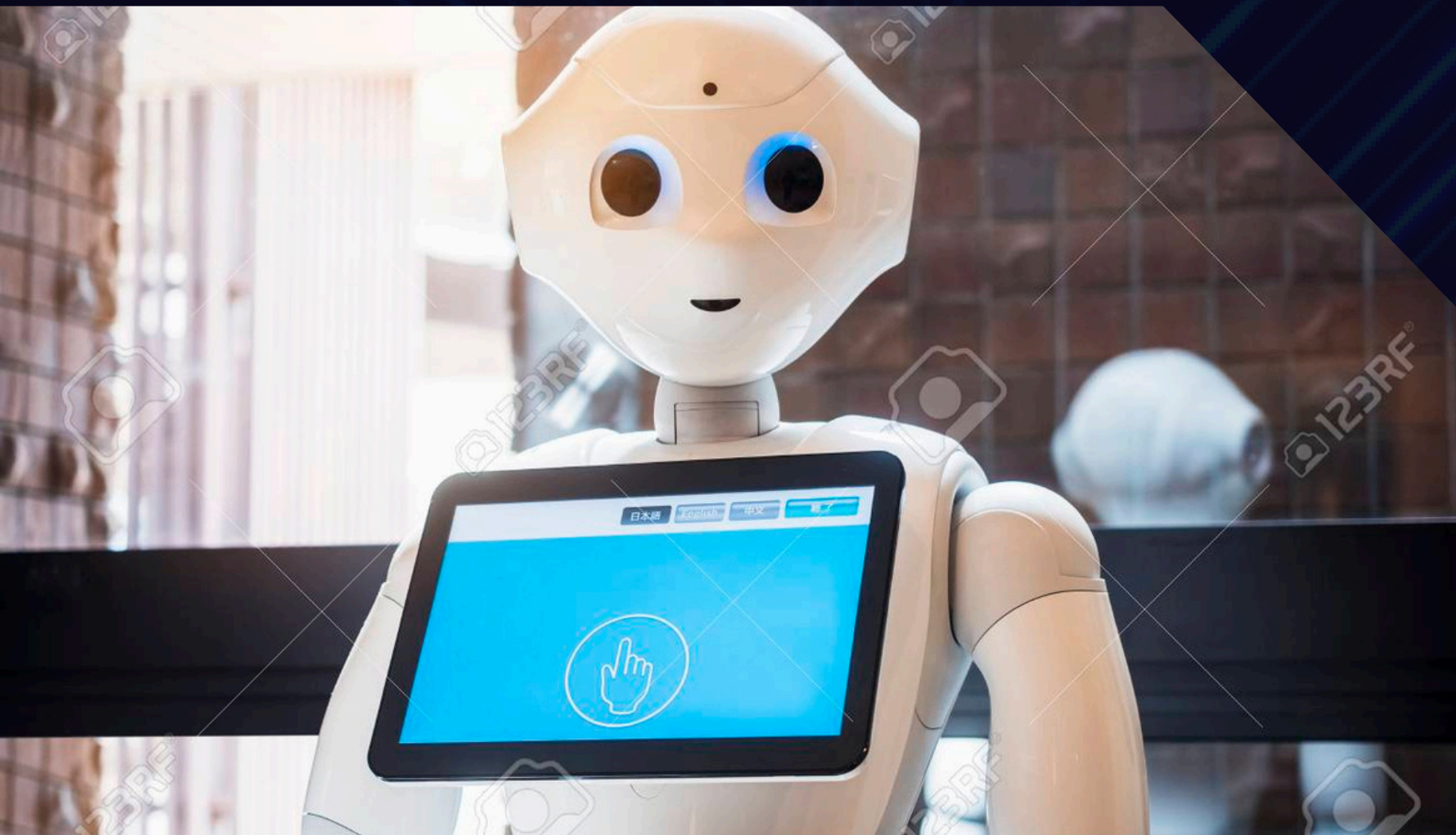
Touchscreen interface for self-guided exploration

Multimedia Experience

Images, videos, 3D models, and real-time captions

Live Map Visualization

Displays current location and navigation path



FEATURES

Exhibits Information Presentation

delivers rich, engaging, and context-aware information about exhibits.

Exhibit Localization & Spatial Awareness

understands where artifacts are and links information to physical locations.



FEATURES



Accessibility Suite

Accessibility Suite

Analytics & Smart Insights

Tracks visitor behavior, questions, and engagement metrics

System Integration & Connectivity

CMS, ticketing, cloud sync, offline mode, and multi-robot coordination

FEATURES

Adaptive Interaction Modes

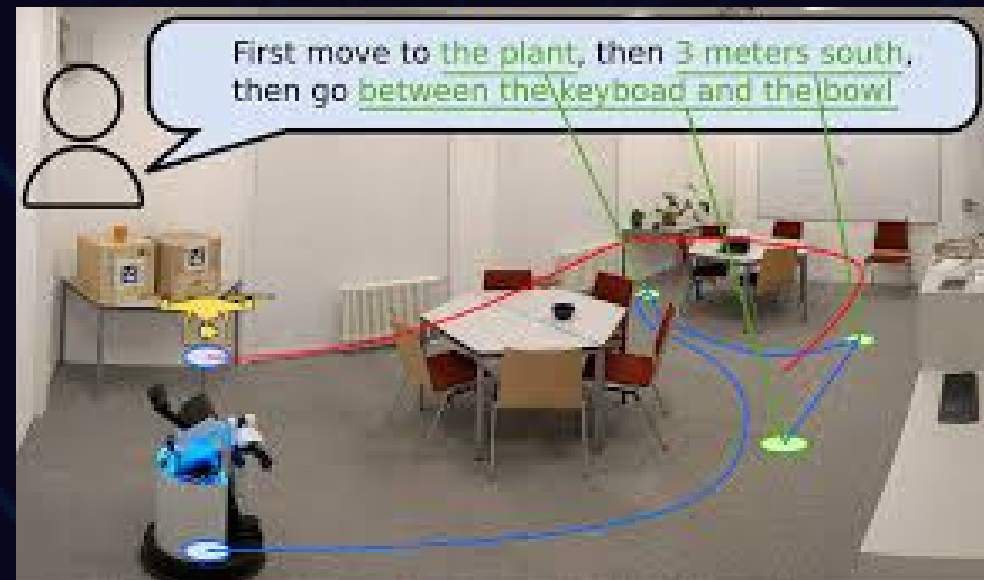
Children mode, expert mode, and sentiment-aware responses

Computer Vision Intelligence

Detects visitors, gestures, faces, and crowd density

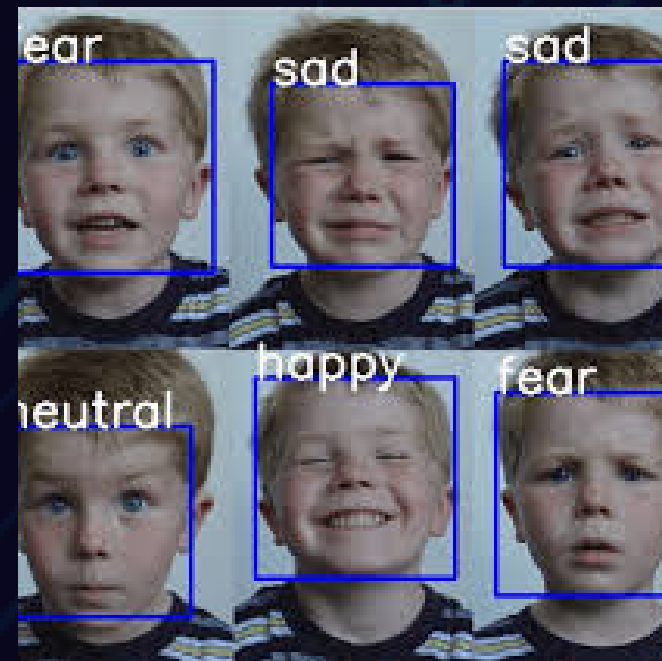


SUBSYSTEMS



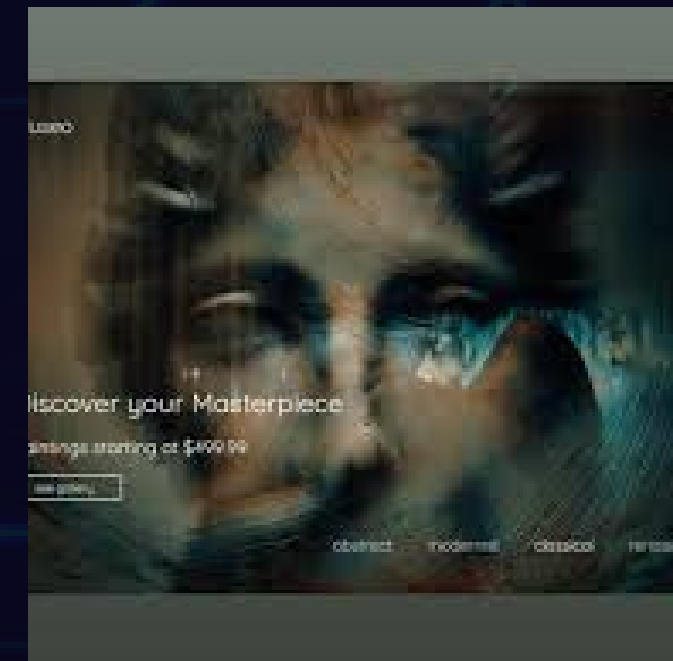
Navigation and Control

Handles the robot's movement and interaction with the environment. It includes path planning, localization, motor control, and real-time obstacle avoidance to ensure smooth and safe navigation.



AI and ComputerVision

Enables the robot to understand and interact intelligently with users and its surroundings. It includes computer vision for detection and recognition, and an LLM-powered system for natural language interaction, question answering, and contextual decision-making.



GUI and Multimedia

Provides the user interface of the robot, displaying exhibit information, playing audio/visual content, and enabling intuitive interaction through a clear and engaging interface.



COMMERCIAL SOLUTION : TEMI

- Price : **\$4999(wifi)**
\$5999(5G)
- General purpose assistant could be customized minimally to act as guide

WHATS NEXT?

ROS2

MicroRos

Nav2

Flask framework

OpenAI -Chatgpt
APi

April tags

